Cover Letter — Ear and Hearing

Date: August 11, 2026

Dear Editor-in-Chief,

We are pleased to submit our manuscript, "A Fuzzy Logic Framework for Sensorineural Hearing Loss Classification: Preserving Diagnostic Gradation Lost to Crisp Thresholds," for consideration as a research article in *Ear and Hearing*.

Hearing loss classification has rested on the same crisp decibel thresholds for over half a century. Those thresholds are practical for surveillance and triage, but they force a continuous biological variable into discrete bins, and the patients who fall within a few decibels of a boundary are the ones most likely to be mislabelled. We developed a Mamdani-type fuzzy inference system that maps pure-tone audiogram thresholds into graded severity vectors, preserving the frequency-specific and borderline information that crisp classification discards. Validated on 5,147 NHANES participants, the system agreed closely with WHO PTA-4 classification (weighted $\kappa = 0.89$) while outperforming it in the borderline zone (91.2% vs 68.4%, $p < 0.01$), where the interpretable rule base gives clinicians a graded picture instead of a binary verdict.

We believe this work fits *Ear and Hearing* particularly well. The journal has consistently published methodological advances in audiological assessment, and the fuzzy framework we present is interpretable by design, which matters for clinical adoption. The analysis code is openly available on GitHub (https://github.com/ameye/fuzzy-audiogram) and the underlying NHANES data are public, supporting reproducibility.

The manuscript reports original research. It has not been published previously and is not under consideration by any other journal. All authors have read and approved the manuscript and declare no conflicts of interest.

We look forward to your consideration. Thank you for your time.

Sincerely,

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